

Introduction to HTML & HTML Elements

What is HTML?

HTML (HyperText Markup Language) is the standard markup language used to create and structure web pages.

HTML is **not a programming language**. It is a **markup language** that tells the browser how content should be displayed.

Examples of content:

- Headings
 - Paragraphs
 - Images
 - Links
 - Tables
 - Forms
 - Videos
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Full Form of HTML

H - Hyper

T - Text

M - Markup

L - Language

What is HyperText?

HyperText is text that contains links to other web pages.

Example:

```
<a href="https://google.com">Open Google</a>
```

Clicking the link navigates the user to another webpage.

What is Markup?

Markup means using **tags** to describe the structure of a webpage.

Example:

```
<h1>This is a Heading</h1>
```

```
<p>This is a paragraph.</p>
```

The browser understands these tags and displays the content accordingly.

Features of HTML

- Easy to learn
 - Platform independent
 - Supported by all browsers
 - Free and open standard
 - Used to build web pages
 - Can integrate with CSS and JavaScript
 - Supports multimedia (audio, video, images)
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Applications of HTML

HTML is used for:

- Creating websites
 - Web applications
 - Landing pages
 - Online forms
 - Blogs
 - E-commerce websites
 - Portfolio websites
 - Educational websites
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Advantages of HTML

- Simple syntax
 - Easy to understand
 - Fast loading
 - Browser compatible
 - SEO-friendly
 - Supports multimedia
 - Works with CSS and JavaScript
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Limitations of HTML

- Cannot perform calculations
- Cannot store data permanently
- Cannot interact with databases
- Cannot create dynamic applications alone

- Needs CSS for styling
 - Needs JavaScript for interactivity
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History of HTML

Version	Year
HTML 1.0	1993
HTML 2.0	1995
HTML 3.2	1997
HTML 4.01	1999
XHTML	2000
HTML5	2014 (Current Standard)

HTML5 Features

- Semantic elements
 - Audio support
 - Video support
 - Canvas
 - SVG
 - Local Storage
 - Geolocation
 - Drag and Drop
 - Better Forms
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Structure of an HTML Document

<!DOCTYPE html>

<html>

<head>

<title>My First Page</title>

</head>

<body>

<h1>Welcome</h1>

<p>This is my first webpage.</p>

</body>

</html>

Explanation

1. <!DOCTYPE html>

Declares the document as HTML5.

2. <html>

Root element of the webpage.

Everything is written inside this tag.

3. <head>

Contains information about the webpage.

Examples:

- Title
 - CSS files
 - Meta tags
 - Icons
-

4. <title>

Displays the webpage title on the browser tab.

Example:

<title>My Website</title>

5. <body>

Contains all visible content.

Example:

```
<body>
```

```
<h1>Hello</h1>
```

```
<p>Welcome</p>
```

```
</body>
```

What is an HTML Element?

An HTML element consists of:

- Opening tag
- Content
- Closing tag

Example:

```
<p>This is a paragraph.</p>
```

Here,

Opening Tag

```
<p>
```

Content

This is a paragraph.

Closing Tag

```
</p>
```

General Syntax

```
<tagname>Content</tagname>
```

Example

```
<h1>Heading</h1>
```

```
<p>Paragraph</p>
```

HTML Tag vs HTML Element

HTML Tag

Only the tag.

Example:

```
<p>
```

or

```
</p>
```

HTML Element

Opening tag + Content + Closing tag.

Example:

```
<p>Hello World</p>
```

Types of HTML Elements

1. Container Element

Has both opening and closing tags.

Example:

```
<h1>Welcome</h1>
```

```
<p>Hello</p>
```

```
<div>Content</div>
```

2. Empty Element (Void Element)

Has no closing tag.

Examples:

```
<br>
```

```
<hr>
```

```
<img>
```

`<input>`

`<link>`

Example:

``

`<br>`

`<hr>`

Nested Elements

An element inside another element is called a nested element.

Example:

`<div>`

`<h1>Heading</h1>`

`<p>Paragraph</p>`

`</div>`

HTML Attributes

Attributes provide additional information about an element.

Syntax:

`<tagname attribute="value">`

Example:

``

Here,

- `src` → image location
- `alt` → alternative text

Another example:

```
<a href="https://google.com">Google</a>
```

href is an attribute.

Common HTML Attributes

Attribute	Description
id	Unique identifier
class	Group elements
style	Inline CSS
title	Tooltip text
href	Hyperlink destination
src	File location
alt	Alternative text
width	Width
height	Height
name	Element name
value	Default value

HTML Comments

Comments are ignored by the browser and are used to document code.

Syntax:

```
<!-- This is a comment -->
```

Example:

```
<!-- Main Heading -->
```

```
<h1>Welcome</h1>
```

Basic HTML Example

```
<!DOCTYPE html>

<html>

  <head>

    <title>Student Portal</title>

  </head>

  <body>

    <h1>Welcome to HTML</h1>

    <p>HTML is used to create webpages.</p>

    <a href="https://google.com">
      Visit Google
    </a>

    <br><br>

    

  </body>

</html>
```

Browser Output

Welcome to HTML

HTML is used to create webpages.

Visit Google

Best Practices

- Always use `<!DOCTYPE html>`.
 - Close all non-void elements properly.
 - Use lowercase tag names.
 - Indent code consistently for readability.
 - Add meaningful alt text for images.
 - Use semantic HTML elements where appropriate.
 - Validate HTML to avoid errors.
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Summary

- HTML stands for **HyperText Markup Language**.
 - HTML structures web pages using **elements** and **tags**.
 - An HTML document contains `<!DOCTYPE>`, `<html>`, `<head>`, and `<body>`.
 - HTML elements are made up of an opening tag, content, and a closing tag.
 - Attributes provide extra information to elements.
 - HTML5 is the current standard and supports modern web development features.
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